

Quantum Error Correction

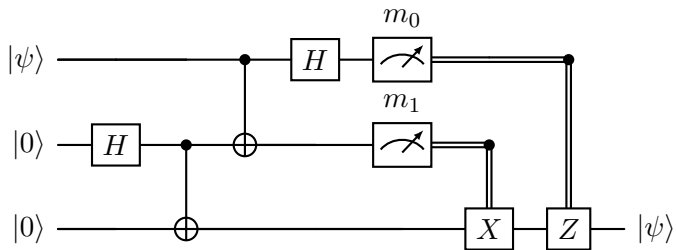
Part 2: Clifford Hierarchy

{Akash Bhople, Qingyan Hu, Tate Middleton, Thomas Wang, Linfeng Wei, Kingsley Yeon, Bobby Zhang} $\subseteq$ **Clifford Hierarchy** $\subseteq$ **QEC**

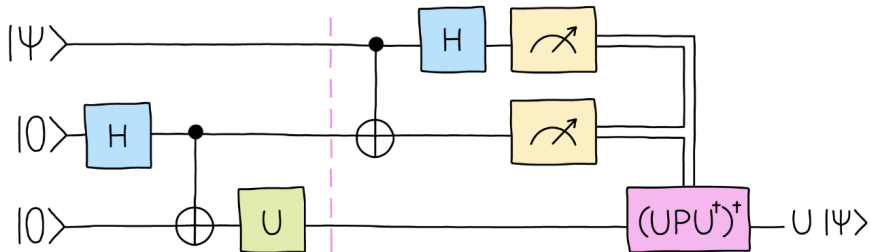
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Why Clifford hierarchy?

- Universal quantum computation via teleportation [Gottesman & Chuang, 1999]

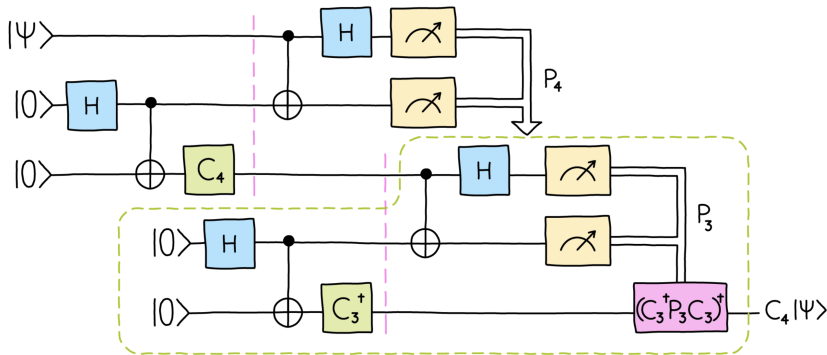


- What about teleporting a gate?

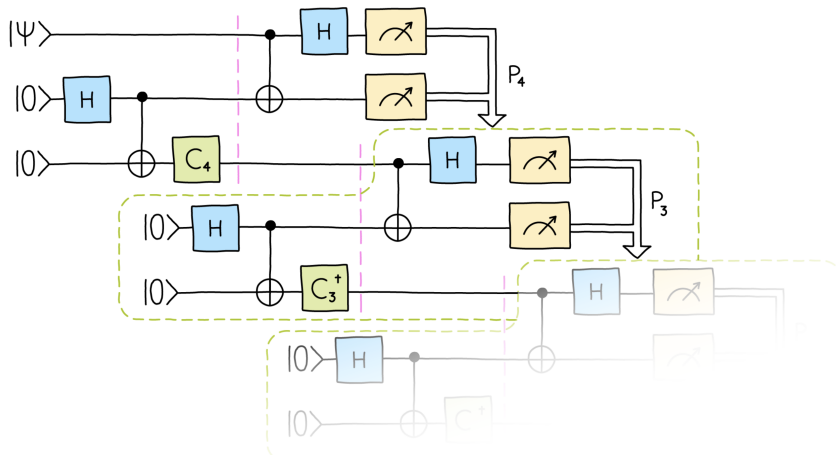


Definition

Let $\mathcal{C}_1 = \mathcal{P}_n$, the group of n -qubit Paulis. The *Clifford hierarchy* is recursively defined by

$$\mathcal{C}_{k+1} = \{U \in \mathbb{U}(2^n) : U\mathcal{P}_n U^\dagger \subseteq \mathcal{C}_k\}.$$


- We can keep going...



Semi-Clifford gates

- $\mathcal{C}_k$'s have wild structures:
 - ▶ $\mathcal{C}_1$: Pauli group, $\mathcal{C}_2$ Clifford group
 - ▶ For $k \geq 3$, $\mathcal{C}_k$ is *not* a group

Definition

- $U \in \mathbb{U}(2^n)$ is **semi-Clifford** if $U = L_1 D L_2$, for some $L_1, L_2 \in \mathcal{C}_2$ and diagonal D .
- $U \in \mathbb{U}(2^n)$ is **generalized semi-Clifford** if $U = L_1 \Pi D L_2$, for some $L_1, L_2 \in \mathcal{C}_2$ and diagonal D .

Theorem

- U is semi-Clifford iff $U\mathcal{G}U^\dagger \subseteq \mathcal{P}_n$ is a maximal abelian subgroup for a maximal abelian subgroup $\mathcal{G} \subseteq \mathcal{P}_n$
- U is generalized semi-Clifford iff $U \text{span}(\mathcal{G})U^\dagger = \text{span}(\mathcal{G}')$ for maximal abelian subgroups $\mathcal{G}, \mathcal{G}' \subseteq \mathcal{P}_n$

- Why semi-Clifford?

Teleportation depth [Zeng, Chen, Chuang, 2008]

How many teleportation steps are necessary, on average, to implement a given gate

	1-Bit Teleportation (Semi-Clifford)	2-Bit Teleportation (General)
Bound $T(n, k)$	$2^n \left[1 - \left(1 - \frac{1}{2^n} \right)^{k-2} \right]$	$4^n \left[1 - \left(1 - \frac{1}{4^n} \right)^{k-2} \right]$
Limit $(k \rightarrow \infty)$	$\rightarrow 2^n$	$\rightarrow 4^n$

- Semi-Cliffords are **efficient**

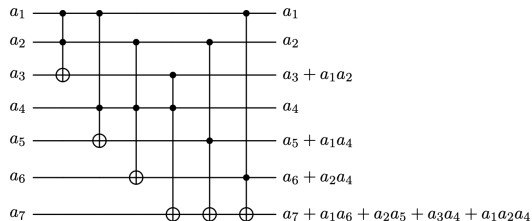
Theorem

- For $n = 1, 2$, **all** $\mathcal{C}_k$ gates are semi-Clifford [Zeng, Chen, Chuang, 2008]
 - For $n = 3$, **all** elements of $\mathcal{C}_3$ are semi-Clifford [Zeng, Chen, Chuang, 2008]
 - For all n : **all** $\mathcal{C}_3$ gates are generalized semi-Clifford [Beigi & Shor, 2009]
 - For $n = 4$, **All** elements of $\mathcal{C}_3$ are semi-Clifford [Anderson, Connelly, 2025]
 - There are gates in $\mathcal{C}_3$ that are **not** semi-Clifford [He, Robitaille, Tan, 2025]
-
- For $n > 2, k > 3$: $\mathcal{C}_k$ strictly contains *non-semi-Clifford* gates.

Example (Zeng, Chen, Chuang, 2008)

Let $V_{3,k} = \text{diag}(1, e^{i\pi/2^{k-1}})$ for $k > 3$ in computational basis. Then $W_k = \text{Toffoli}_{1,2 \rightarrow 3} \cdot V_{3,k} \in \mathcal{C}_{k+1}$ is **not semi-Clifford**.

Permutation gates in the third level



Theorem (He, Robitaille, Tan, 2025)

Every permutation $\Pi \in \mathcal{C}_3$ can be factored as

$$\Pi = P_1 \mu P_2.$$

Clifford permutations P_1, P_2 and a staircase-form permutation. Moreover, Staircase-form $\mathcal{C}_3$ permutations and descending multiplications are in 1-1 correspondence.

¹Figure from [He, Robitaille, Tan, 2025]

Diagonals Clifford hierarchy

- What about some simpler subsets of $\mathcal{C}_\ell$?
- $\{D \in \mathcal{C}_\ell : D \text{ is diagonal}\}$ is a group [Zeng, Chen, Chuang, 2008]
- Every n -qubit diagonal in $\mathcal{C}_\ell$ can be written as [Cui, Gottesman, Krishna, 2016]

$$U^\ell = e^{i\phi} \sum_{x \in \{0,1\}^n} \exp \left[2\pi i \sum_{\emptyset \neq S \subseteq [n], |S| \leq k} \frac{c_S}{2^{k-|S|+1}} \prod_{j \in S} x_j \right] |x\rangle \langle x|$$

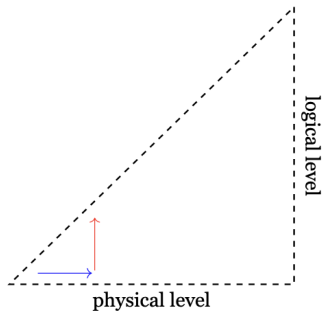
Climbing the Diagonal Hierarchy with CSS Codes [Hu, Liang, Calderbank, 2025]

In $\mathcal{C}_\ell$, we can **(1)** concatenate, **(2)** remove Z -stabilizers, and **(3)** add X -stabilizers to obtain a CSS code carrying a logical $U^{(\ell+1)}$.

Step	Code	Level
1. Concatenate	$[[n, k, d]] \longrightarrow [[2n, k, d' \geq d]]$	$\ell'_{\text{phys}} = \ell_{\text{phys}} + 1, \quad \ell'_{\text{log}} = \ell_{\text{log}}$
2. Remove Z -stab.	$\longrightarrow [[2n, k' \geq k, d'' \leq d']]$	$\ell''_{\text{log}} = \ell_{\text{log}} + 1$
3. Add X -stab.	$\longrightarrow [[2n, k'' \leq k', d''' \geq d'']]$	unchanged

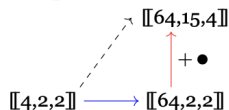
$$\underset{\text{Pauli}}{Z} \xrightarrow{\sqrt{\cdot}} \underset{\text{Clifford}}{S} = \sqrt{Z} \xrightarrow{\sqrt{\cdot}} \underset{\text{Third Level}}{T} = \sqrt{S}.$$

Ideal: TSZ , X error after Z : $TSXZ = T \underbrace{(SXS^\dagger)}_{=Y} SZ = \underbrace{(TYT^\dagger)}_{=(Y-X)/\sqrt{2}} TSZ.$



- Concatenation
- ↑ Removing Z -stabilizers
- Adding X -stabilizers

Example:



¹Figure from [Hu, Liang, Calderbank, 2025]

Climbing the Clifford hierarchy?

- For Hermitian unitary U , define $\hat{U} = e^{i\pi U/4} = \frac{I+iU}{\sqrt{2}}$, so $\hat{U}^2 = iU$.

- **Question:**

$$U \in \mathcal{C}_k \Rightarrow? \hat{U} \in \mathcal{C}_{k+1}$$

- Not always: $H \in \mathcal{C}_2$, but $\hat{H} \notin \mathcal{C}_3$, since $HXH = Z$ and $XZ = -ZX$.

For Hermitian Clifford C , its Pauli action is encoded by

$$CE(v)C^\dagger = \pm E(vF_C), \quad F_C \in \text{Sp}(2n, 2).$$

[Bastioni et al, 2026] proves the classification

$$\hat{C} \in \mathcal{C}_3 \iff F_C \text{ is hyperbolic and } \dim \text{rowspan}(I + F_C) = 2.$$

where hyperbolic means

$$\langle vF_C, v \rangle_s = 0 \quad \forall v \in \mathbb{F}_2^{2n}.$$

Let's Wrap It Up!

Disclaimer: the figures below were generated by an LLM, colour-matched to the speaker's shirt.

The space of codes: what actually determines the level

- How do we do fault-tolerant computation?

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- A gate $U \in \mathbb{U}(2^n)$ is **transversal** if $U = \bigotimes_i U_i$ with $\text{supp}(U_i) \subseteq Q_i$

The space of codes: what actually determines the level

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- Question: what are the highest-level transversal logical operators you can implement?

The space of codes: what actually determines the level

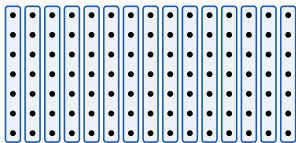
- A **code** is a subspace $\mathcal{Q} \subseteq (\mathbb{C}^2)^{\otimes n}$ with $\dim \mathcal{Q} = 2^k$.

The space of codes: what actually determines the level

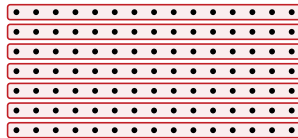
- A **code** is a subspace $\mathcal{Q} \subseteq (\mathbb{C}^2)^{\otimes n}$ with $\dim \mathcal{Q} = 2^k$.
- A **partition** is a grouping $[n] = Q_1 \sqcup \cdots \sqcup Q_N$

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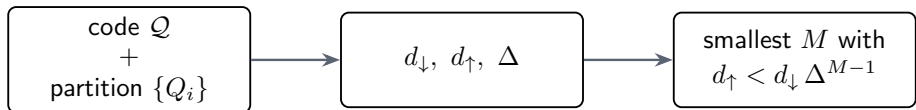
fifteen blocks of seven
saturated by transversal T



seven blocks of fifteen
saturated by transversal H

The space of codes: what actually determines the level

- Question: what are the highest-level transversal logical operators you can implement?

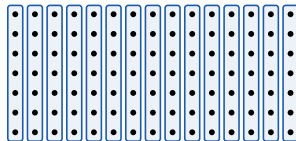


$\Rightarrow$ every transversal logical gate lie in $\bar{\mathcal{C}}_M$

The space of codes: fix code, vary partition

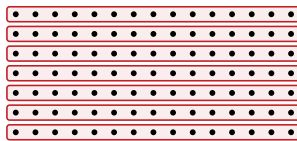
$[[105, 1, 9]]$: quantum Reed–Muller $[[15, 1, 3]]$ with each qubit encoded in Steane $[[7, 1, 3]]$.

$$M = 3$$



fifteen blocks of seven
blocks weigh as the $[[15, 1, 3]]$
 $d_{\downarrow} = 3$, $d_{\uparrow} = 7$, $\Delta = 15/7$
transversal T

$$M = 2$$



seven blocks of fifteen
blocks weigh as the $[[7, 1, 3]]$
 $d_{\downarrow} = d_{\uparrow} = 3$, $\Delta = 7/3$
transversal H

The space of codes: vary code, fix partition

All four with the **singleton partition** $Q_i = \{i\}$

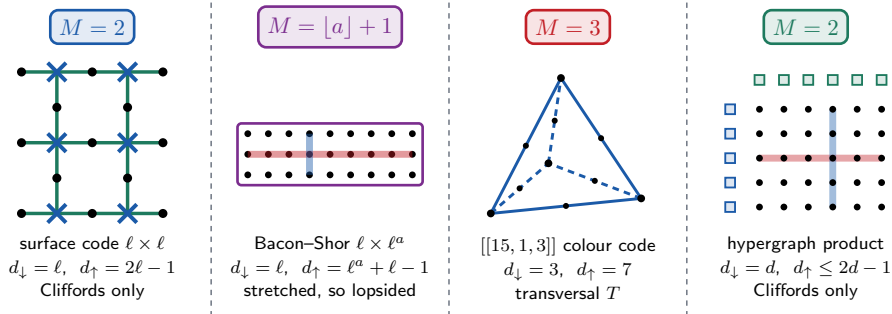
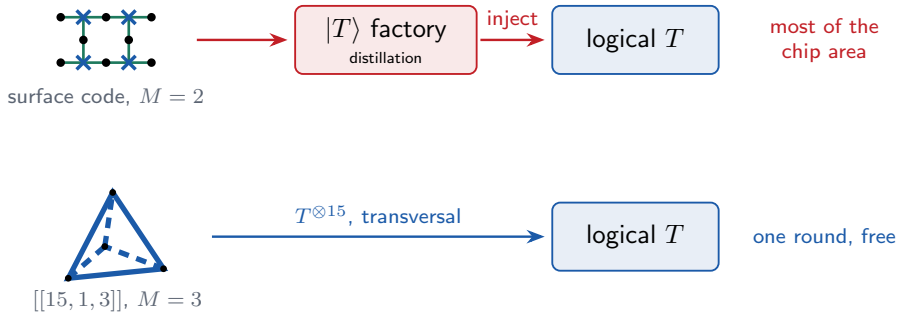


Figure generated with Claude.

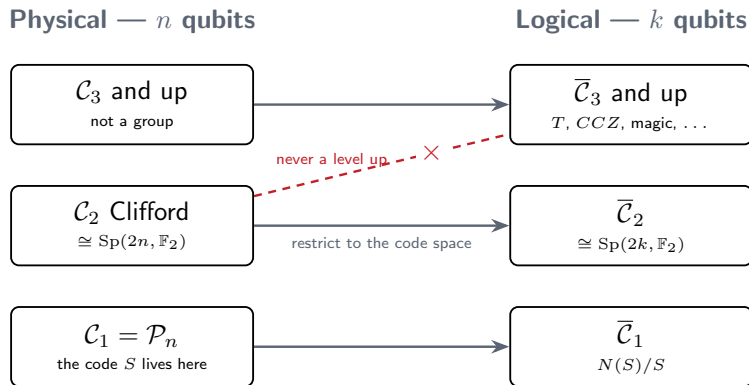
Why level three?



Why go above level three?

- **Multi-controlled gates.** $C^w Z$ sits at level $w + 1$ (CCZ and Toffoli at level 3, $CCCZ$ at level 4).
- **Dyadic rotations.** $\text{diag}(1, e^{i\pi/2^{k-1}})$ is native at level k (level 1 is Z , level 2 is S , level 3 is T), but otherwise costs $\Theta(\log(1/\varepsilon))$ T gates to synthesise.
- **Quantum Fourier transform.** Controlling a level- k rotation gate costs one level up (CS is 3, CT is 4).
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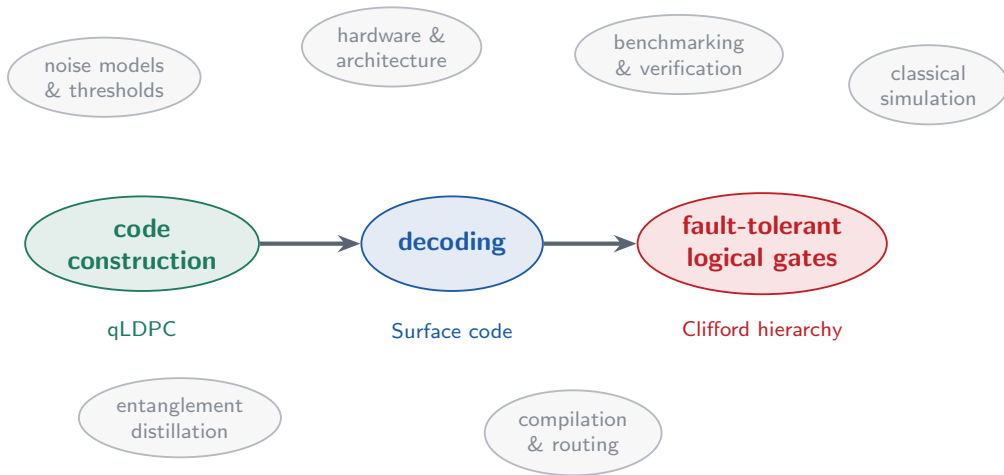
Physical $\rightarrow$ logical



A physical U preserving the code space induces $\bar{U}$ on k qubits, and $U \in \mathcal{C}_j \Rightarrow \bar{U} \in \bar{\mathcal{C}}_j$.

Mechanism	Idea	Cost	Leaves $\bar{C}_M$?
Free, code unchanged			
Transversal gates	$U = \bigotimes_i U_i$	none	no
Code automorphisms	permutations preserving S	a relabelling	no
Fold-transversal gates	a duality of the Tanner graph	a relabelling	no
With cost, code unchanged			
Gate teleportation & magic states	non-Clifford resource moved offline	factory area	yes
Pieceable fault tolerance	interleave EC rounds inside a non-transversal circuit	time	yes
Lattice surgery	merge and split patches	patch area, time	no
With cost, code changed			
Code switching, gauge fixing	two codes, complementary transversal sets	switching protocol	yes
Concatenation, climbing constructions	engineer a higher logical level	qubit overhead	yes

Summary



Thank You!

Questions?

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Zhang} $\subseteq$ **QEC**